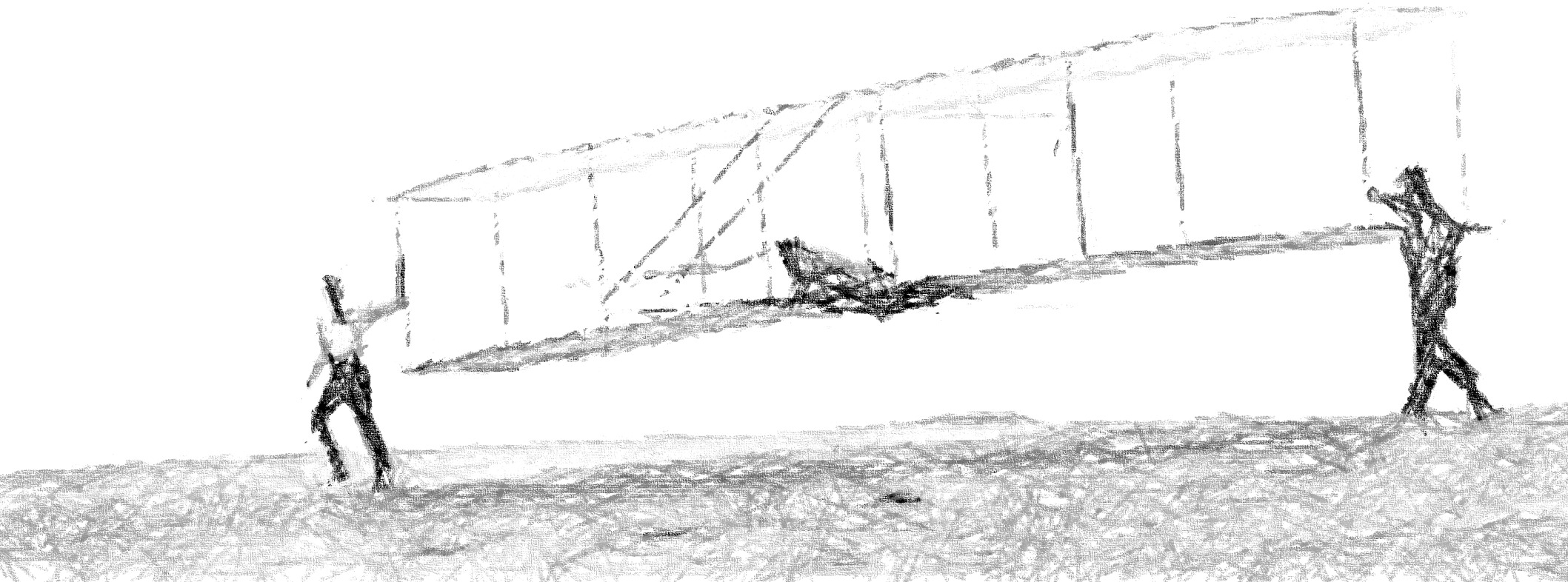




Mechanics applied to Aerospace Engineering

Course 251-14165 – BSc Aerospace Engineering

2. Particle dynamics



Contents

- Mass, linear momentum and kinetic energy
- Newton's laws of motion
- Internal and external forces
- Conservation of linear momentum
- Equations of motion of a point particle
- Power, work, and potential energy
- Conservation of mechanical energy
- Equilibrium conditions



Mass, momentum and energy

- Point particles have a physical property known as mass, m^P
 - The mass of a point particle is constant
- The (linear) momentum of a point particle P is defined as

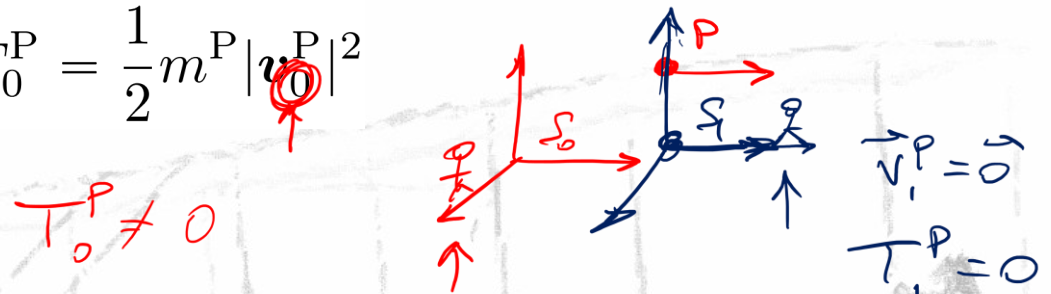
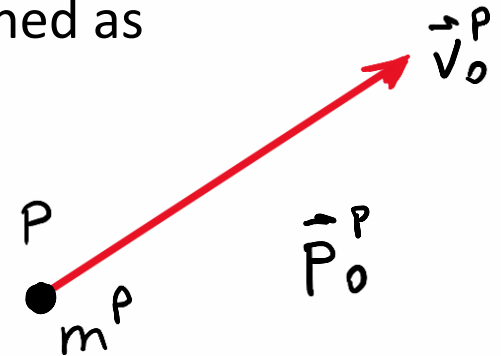
$$\mathbf{p}_0^P = m^P \mathbf{v}_0^P$$

- It is a vector quantity
- Units: ML/T (kg m/s)

- The kinetic energy of a point particle P is

$$T_0^P = \frac{1}{2} m^P |\mathbf{v}_0^P|^2$$

- It is a scalar quantity
- Units: ML²/T² (kg m²/s²)



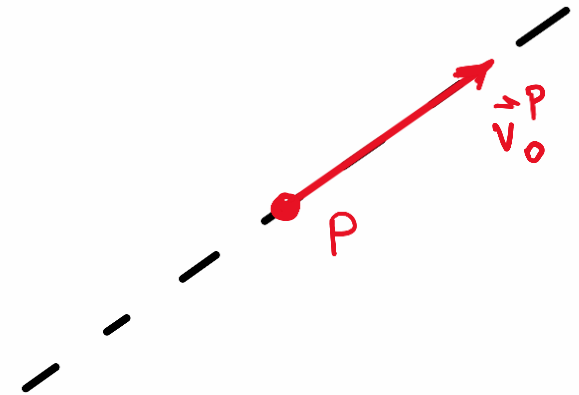
As both $\mathbf{p}_0^P$ and T_0^P depend on the particle's velocity, they depend on the reference frame from where they are computed.

Newton's laws of motion

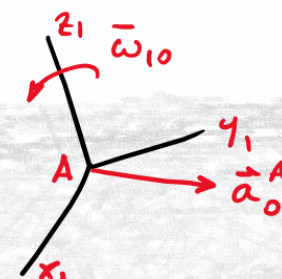
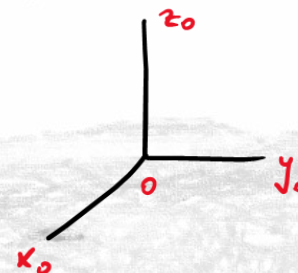
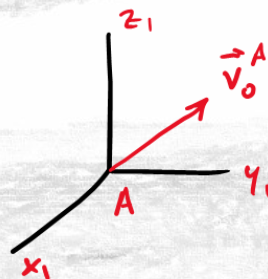
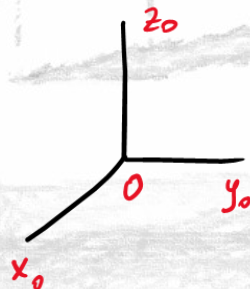
- First law: *A particle shall continue in a state of rest or of uniform motion in a straight line unless impressed upon by a force.*

- This innocent-looking statement can be turned around and regarded as the *definition* of a special class of reference frames: *inertial reference frames*

A reference frame is *inertial* if any free point particle remains at rest or travels in uniform rectilinear motion in the absence of forces



- Newton's laws are only satisfied in inertial reference frames. The first law sets this condition implicitly
- We shall take as an axiom the existence of at least one inertial frame
 - Given one inertial frame, any other frame not rotating and not accelerating with respect to it is also inertial
 - Not all reference frames are inertial! in fact, most are not.

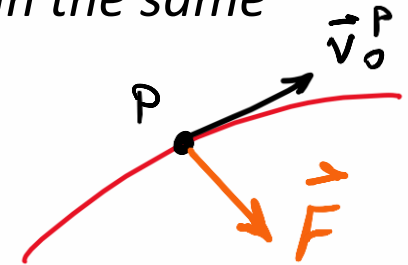


Newton's laws of motion

- Second law: *The rate of change of linear momentum of a particle is proportional to the net impressed force and takes place in the same direction as the force*

- Mathematically:

$$\left. \frac{d\mathbf{p}_0^P}{dt} \right|_0 = m^P \left. \frac{d\mathbf{v}_0^P}{dt} \right|_0 = m^P \mathbf{a}_0^P = \mathbf{F}$$



- Remember: this is only valid in an inertial reference frame

- A *force* $\mathbf{F}$ is a vector quantity that represents an interaction between “something else” and our particle. It affects the motion of the particle

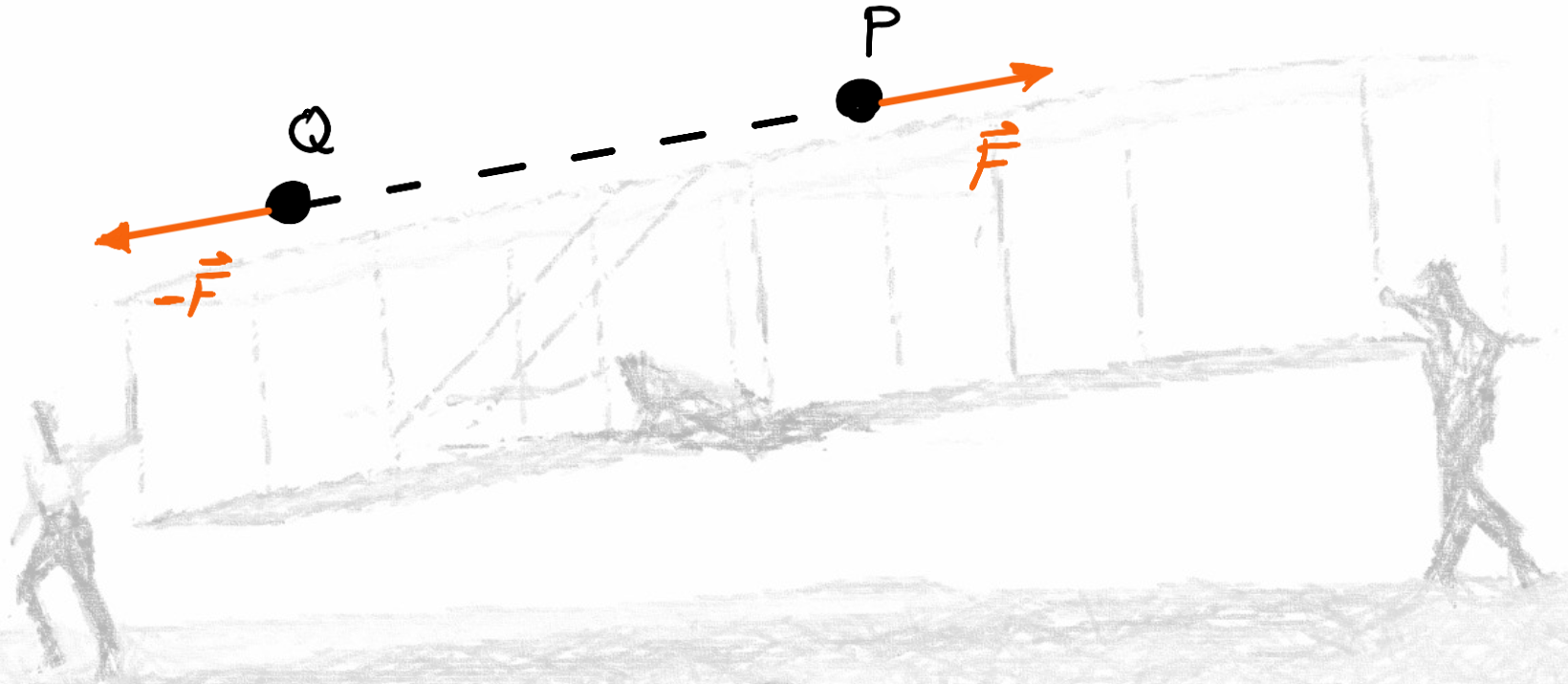
- Units: ML/T² (N)

- The second law establishes how the velocity of a point particle changes in time. It embodies the *equations of motion*, or the *dynamics*, of the particle

- This variation is proportional to the vector sum of all forces acting on the particle. Thus, a larger net force results in a larger acceleration
- The variation is inversely proportional to the particle's mass (its *inertia*). Thus, the more massive the particle, the slower it will accelerate for a given force.

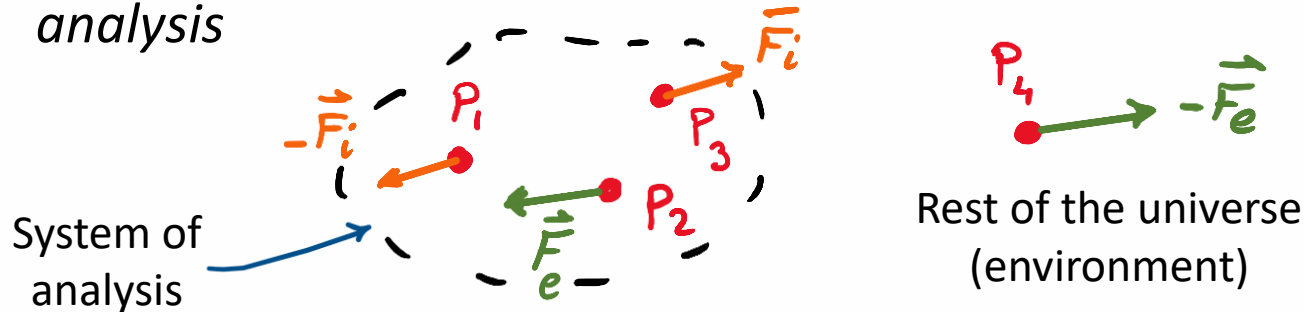
Newton's laws of motion

- Third law: *To every action there is an equal and opposite reaction*
 - In other words: if a particle Q exerts a force $\mathbf{F}$ on a particle P, then the particle P must exert a force $-\mathbf{F}$ on particle Q
 - The terms action and reaction are used to refer to these two forces
 - Moreover, the two forces share the same *line of action*.
- The third law forbids the existence of unpaired forces in the universe!



Internal and external forces

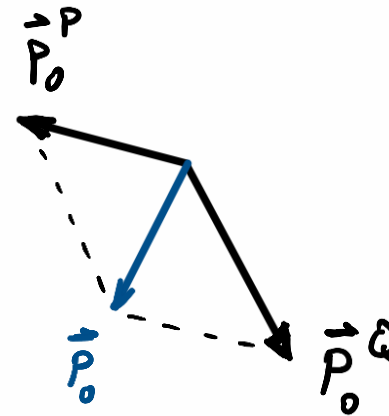
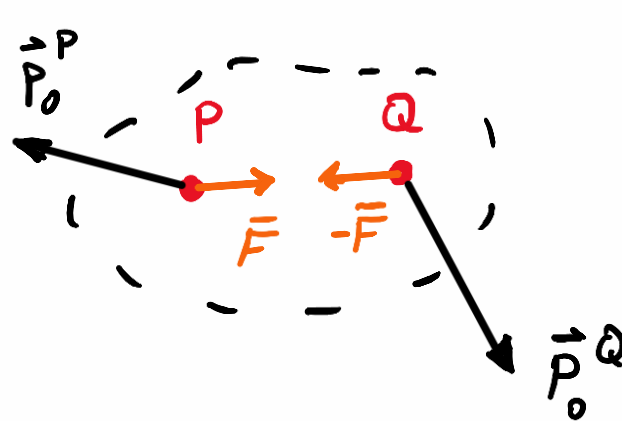
- In mechanics, we can consider the total linear momentum of a group of particles simultaneously. We refer to those particles as our *system of analysis*



- *Internal forces* are those among the particles in the system of analysis. Then, the action and reaction forces are both within the system.
 - In other words, the cause of an internal force on a particle of the system is inside the system, and its reaction is felt by another particle in the system
- *External forces* are those whose cause is outside the system of analysis, i.e., in the *environment*. The reaction associated to an external force is felt somewhere else in the universe, but not on any of the particles of the system.
- We say that a system of analysis is *isolated* if there are no external forces acting on its particles

Conservation of linear momentum

- The third law conceals an important principle in mechanics:
conservation of total linear momentum $\mathbf{p}_0 = \mathbf{p}_0^P + \mathbf{p}_0^Q$:



$$\left. \frac{d\mathbf{p}_0^P}{dt} \right|_0 = \mathbf{F},$$

$$\left. \frac{d\mathbf{p}_0^Q}{dt} \right|_0 = -\mathbf{F}.$$

$$\left. \frac{d(\mathbf{p}_0^P + \mathbf{p}_0^Q)}{dt} \right|_0 = \mathbf{0} \Rightarrow \mathbf{p}_0^P + \mathbf{p}_0^Q = \text{const}$$

- The total linear momentum of an isolated system is constant

Equations of motion

- The second law states that the change in momentum per unit time of a particle equals the sum of all forces acting on it.
 - This describes the *evolution* or the *dynamics* of the point particle

$$m^P \mathbf{a}_0^P = \mathbf{F}$$

- This is a vector equation that can be used to obtain three scalar *second-order ordinary differential equations* for the time evolution of the coordinates used to describe the position of the particle
 - To do so, we *project* the vector equation along three linearly-independent vector directions of our choice.
 - For example, if our coordinates are x, y, z , projecting along $\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0$ yields:

$$\frac{d^2x}{dt^2} = F_x/m^P, \quad \frac{d^2y}{dt^2} = F_y/m^P, \quad \frac{d^2z}{dt^2} = F_z/m^P$$

Choosing wisely the projection vectors (or looking for linear combinations of the resulting scalar equations) can result in simpler, more compact, or more insightful expressions

Equations of motion

$$\frac{d^2x}{dt^2} = F_x/m^P, \quad \frac{d^2y}{dt^2} = F_y/m^P, \quad \frac{d^2z}{dt^2} = F_z/m^P$$

- These *equations of motion* can be alternatively written as 6 first-order ordinary differential equations for $x, \dot{x}, y, \dot{y}, z, \dot{z}$:

$$\begin{aligned} \frac{d\dot{x}}{dt} &= F_x/m^P; & \frac{d\dot{y}}{dt} &= F_y/m^P; & \frac{d\dot{z}}{dt} &= F_z/m^P; \\ \frac{dx}{dt} &= \dot{x}; & \frac{dy}{dt} &= \dot{y}; & \frac{dz}{dt} &= \dot{z}, \end{aligned}$$

- These equations can be integrated to yield the trajectory of the point particle as a function of time

- The solution will depend on 6 integration constants

$$x = x(t; C_1, C_2, C_3, C_4, C_5, C_6),$$

$$y = y(t; C_1, C_2, C_3, C_4, C_5, C_6),$$

$$z = z(t; C_1, C_2, C_3, C_4, C_5, C_6).$$

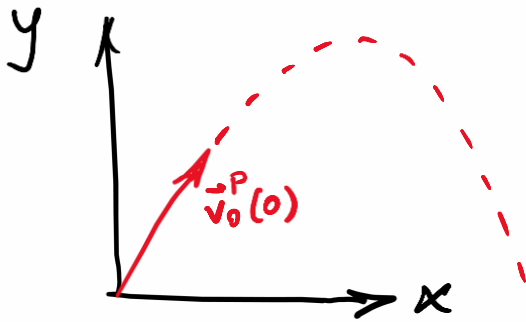
This is what we mean when we say we have *solved the dynamics of P*

- The constants must be determined from the *initial conditions* of the problem: the position and velocity at $t = 0$:

$$\mathbf{r}_0^P(0) \quad \mathbf{v}_0^P(0)$$

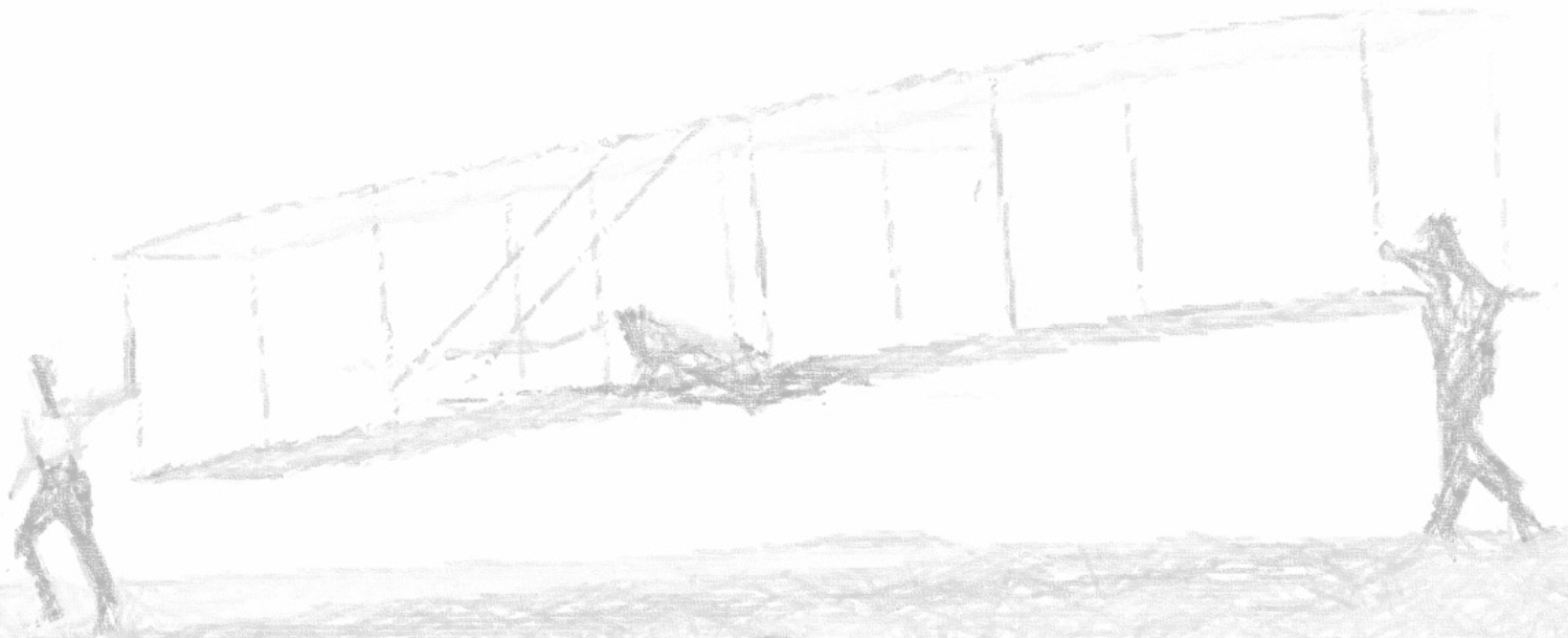
Example

- Integrate the motion of a point particle P with mass $m^P = 1$ kg under the effect of gravity
 - Weight force: $\mathbf{F} = -m^P g \mathbf{j}$ (where $g = 10$ m/s²)
 - Initial conditions: $x(0) = y(0) = 0$, $\dot{x}(0) = 2$ m/s, $\dot{y}(0) = 5$ m/s



Example

- The equations of motion can obviously be used as well to compute the force that would be necessary for a particle to describe a given trajectory.
 - What is the force needed for P ($m^P = 2 \text{ kg}$) to have a velocity $\mathbf{v}_0^P(t) = 2at^2\mathbf{i}_0 + \exp bt\mathbf{j}_0$?



Power and work associated to a force

- The *power* exerted by a force $\mathbf{F}$ at a given instant is

$$\dot{W}_0 = \mathbf{F} \cdot \mathbf{v}_0^P$$

- This power determines the rate of change of the kinetic energy of P:

$$\frac{dT_0}{dt} = \frac{d}{dt} \left(\frac{1}{2} m \mathbf{v}_0^P \cdot \mathbf{v}_0^P \right) = m \mathbf{a}_0^P \cdot \mathbf{v}_0^P = \mathbf{F} \cdot \mathbf{v}_0^P = \dot{W}_0$$

- The time integral of power is the *work* done by the force

- It is a function of the initial and final times of integration, t_1 and t_2 , and in general, the path followed by the particle between these two instants of time:

$$W_0(t_1, t_2) = \int_{t_1}^{t_2} \dot{W}_0 dt = \int_{t_1}^{t_2} \mathbf{F} \cdot \mathbf{v}_0^P dt = \int_{t_1}^{t_2} \mathbf{F} \cdot \left. \frac{d\mathbf{r}_0^P}{dt} \right|_0 dt$$

A force exerts nonzero power and work iff it induces **tangential** acceleration on the particle.

Potential energy of a conservative force

- The work integral can be written as

$$W_0(\mathbf{r}_{0,1}^P, \mathbf{r}_{0,2}^P, \Gamma) = \int_{t_1}^{t_2} \mathbf{F} \cdot \frac{d\mathbf{r}_0^P}{dt} \Big|_0 dt = \int_{\mathbf{r}_{0,1}^P}^{\mathbf{r}_{0,2}^P} \mathbf{F} \cdot d\mathbf{r}_0^P \Big|_0$$

- In general, the integral depends on $\mathbf{r}_{0,1}^P$ and $\mathbf{r}_{0,2}^P$ but also on the path or trajectory Γ followed by the particle
- In some cases, we observe that the result has no dependency on Γ , i.e., the integral yields the same value regardless of the path
 - When this happens, we say $\mathbf{F}$ is *conservative*, and we can define a *potential energy function* by fixing $\mathbf{r}_{0,1}^P$ at an arbitrary position $\mathbf{r}_{0,*}^P$:

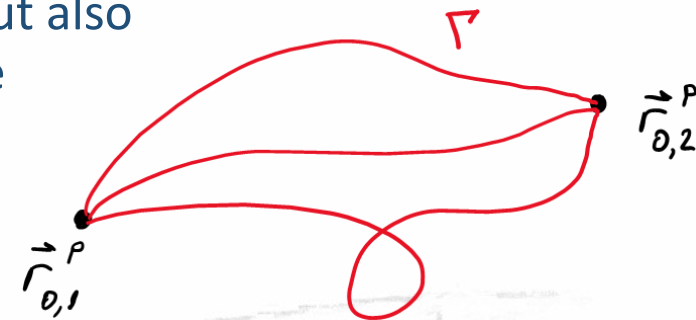
$$V_0(\mathbf{r}_0^P) = - \int_{\mathbf{r}_{0,*}^P}^{\mathbf{r}_0^P} \mathbf{F} \cdot d\mathbf{r}_0^P \Big|_0$$

- The work integral can then be expressed as

$$W_0(\mathbf{r}_{0,1}^P, \mathbf{r}_{0,2}^P) = V_0(\mathbf{r}_{0,1}^P) - V_0(\mathbf{r}_{0,2}^P)$$

- And the force can be expressed as

$$\mathbf{F} = -\nabla V_0$$



The potential energy $V_0(\mathbf{r}_0^P)$ is a scalar function of the position

The “-” sign is just convention

Example

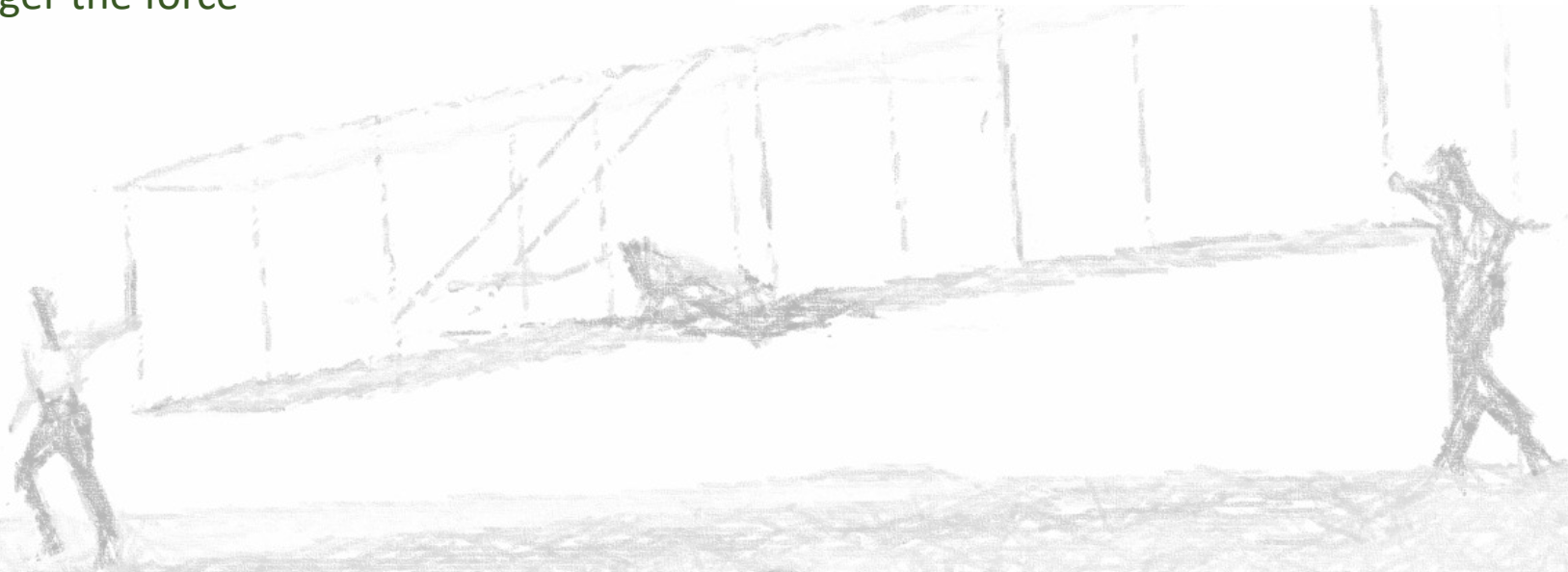
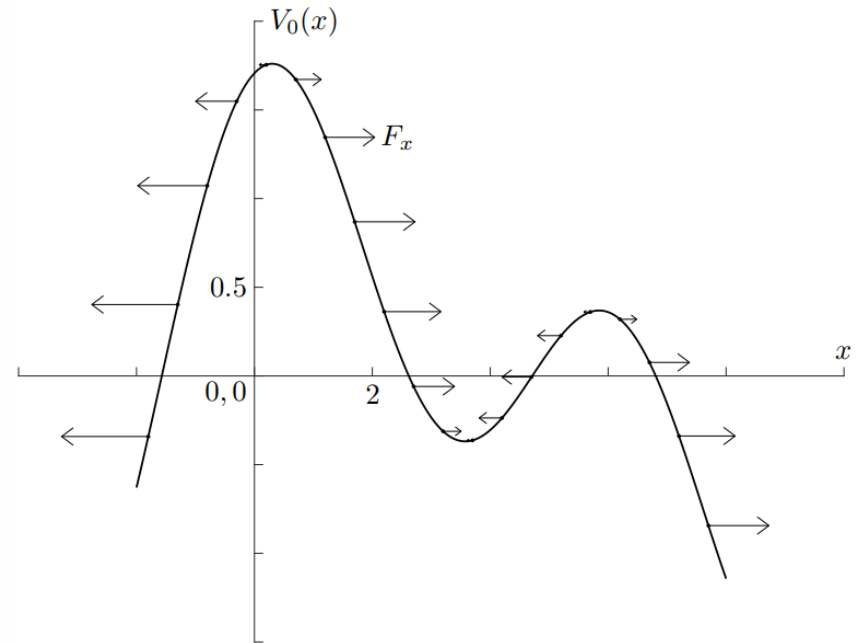
➤ Consider the potential

$$V_0(x) = \cos(x) + \cos(x/2 - \pi/4)$$

■ The force is

$$\mathbf{F} = -\frac{dV_0}{dx} = \sin x + \frac{1}{2} \sin\left(\frac{x}{2} - \frac{\pi}{4}\right)$$

- It pushes the particle *downslope* of the potential (hills and valleys)
- Its magnitude is proportional to the slope. The steeper the potential, the larger the force



Potential energy of a conservative force

- The force is conservative and the potential energy function exists iff:

$$\oint \mathbf{F}(\mathbf{r}_0^P) \cdot d\mathbf{r}_0^P = 0; \quad \forall \Gamma$$

- When the force is differentiable in $\mathbf{r}_0^P$:

If $\nabla \times \mathbf{F} = \mathbf{0}$ in a simply connected domain and $\mathbf{F}$ only depends on $\mathbf{r}_0^P$, then $\mathbf{F}$ is conservative and a potential energy function exists in that domain

- Examples of conservative forces and their potential energies:

- A uniform force like the weight of the particle

$$\mathbf{W} = m^P \mathbf{g}; V_0 = -m^P \mathbf{g} \cdot \mathbf{r}_0^P$$

- The force of a spring of zero natural length attached to the origin

$$\mathbf{F} = -k\mathbf{r}_0^P; V_0 = k(r_0^P)^2/2$$

- A pair of internal forces between P and Q may also be conservative. Then, they have a single associated potential function $V_0 = V_0(\mathbf{r}_0^P, \mathbf{r}_0^Q)$

- Example: electrostatic force on P due to Q:

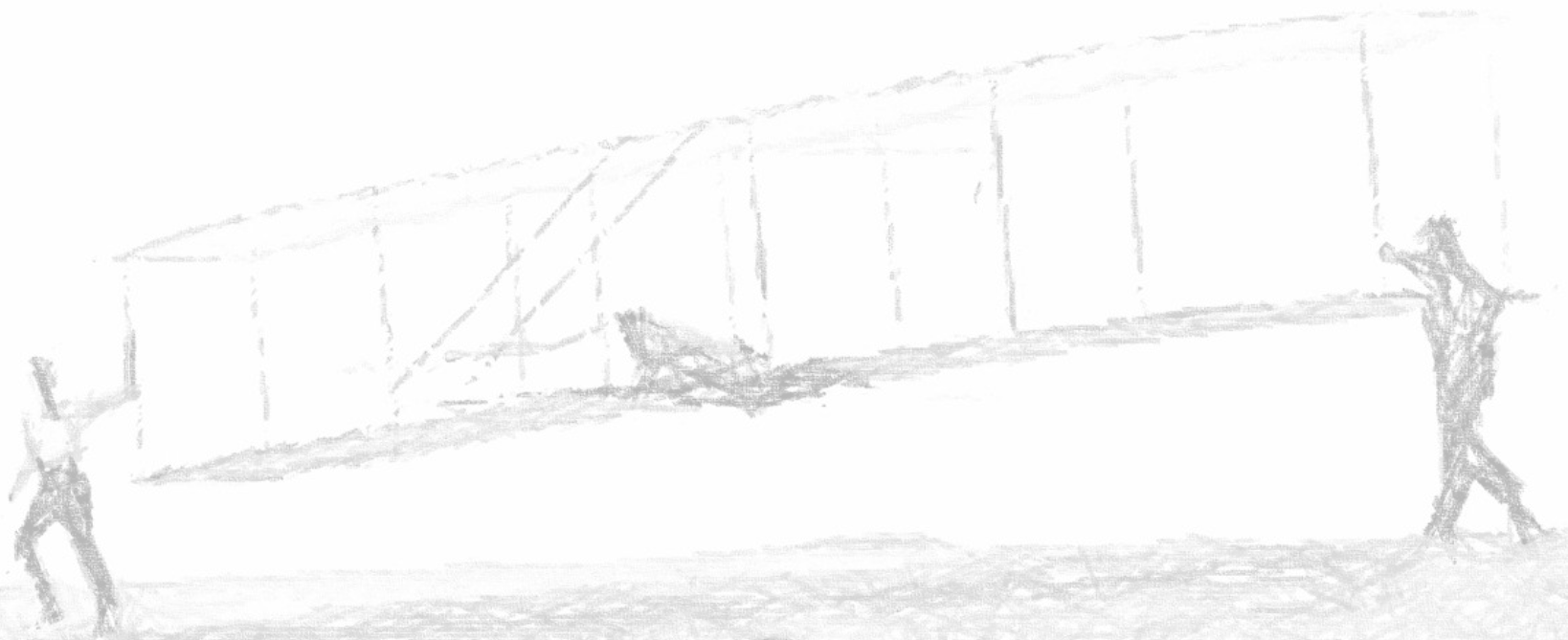
$$\mathbf{F}_P = \frac{k_0 q^P q^Q}{|\mathbf{r}_0^P - \mathbf{r}_0^Q|^3} (\mathbf{r}_0^P - \mathbf{r}_0^Q); \quad V_0 = \frac{k_0 q^P q^Q}{|\mathbf{r}_0^P - \mathbf{r}_0^Q|}$$

$$\mathbf{F}_P = -\frac{\partial V_0}{\partial \mathbf{r}_0^P}$$

$$\mathbf{F}_Q = -\frac{\partial V_0}{\partial \mathbf{r}_0^Q}$$

Examples

- Consider the force $\mathbf{F} = aR\mathbf{e}_R$. Is it conservative? Compute V_0



Examples

- Consider the force $\mathbf{F} = bR^2 \mathbf{e}_\theta$. Is it conservative?



Conservation of mechanical energy

- Consider P subject to a conservative force $\mathbf{F}_C$ and a non-conservative force $\mathbf{F}_N$

$$m^P \left. \frac{d\mathbf{v}_0^P}{dt} \right|_0 = \mathbf{F}_C + \mathbf{F}_N$$

- Dot-multiplying by $\mathbf{v}_0^P$:

$$m^P \left. \frac{d\mathbf{v}_0^P}{dt} \right|_0 \cdot \mathbf{v}_0^P = \frac{1}{2} m^P \frac{d(\mathbf{v}_0^P \cdot \mathbf{v}_0^P)}{dt} = \frac{dT_0^P}{dt}$$

$$\mathbf{F}_C \cdot \mathbf{v}_0^P + \mathbf{F}_N \cdot \mathbf{v}_0^P = -\nabla V_0 \cdot \left. \frac{d\mathbf{r}_0^P}{dt} \right|_0 + \mathbf{F}_N \cdot \mathbf{v}_0^P = -\frac{dV_0}{dt} + \mathbf{F}_N \cdot \mathbf{v}_0^P$$

- We define the mechanical energy as $E_0^P = T_0^P + V_0$:

$$\frac{dT_0^P}{dt} + \frac{dV_0}{dt} = \frac{dE_0^P}{dt} = \mathbf{F}_N \cdot \mathbf{v}_0^P$$

- The mechanical energy of P is constant if the right-hand side is zero. This happens iff:
 - There are no non-conservative forces acting on P, or:
 - The non-conservative forces are always $\perp$ to $\mathbf{v}_0^P$ (i.e., they do no work)

The equation of conservation of energy is a *first integral of motion* of the system of analysis

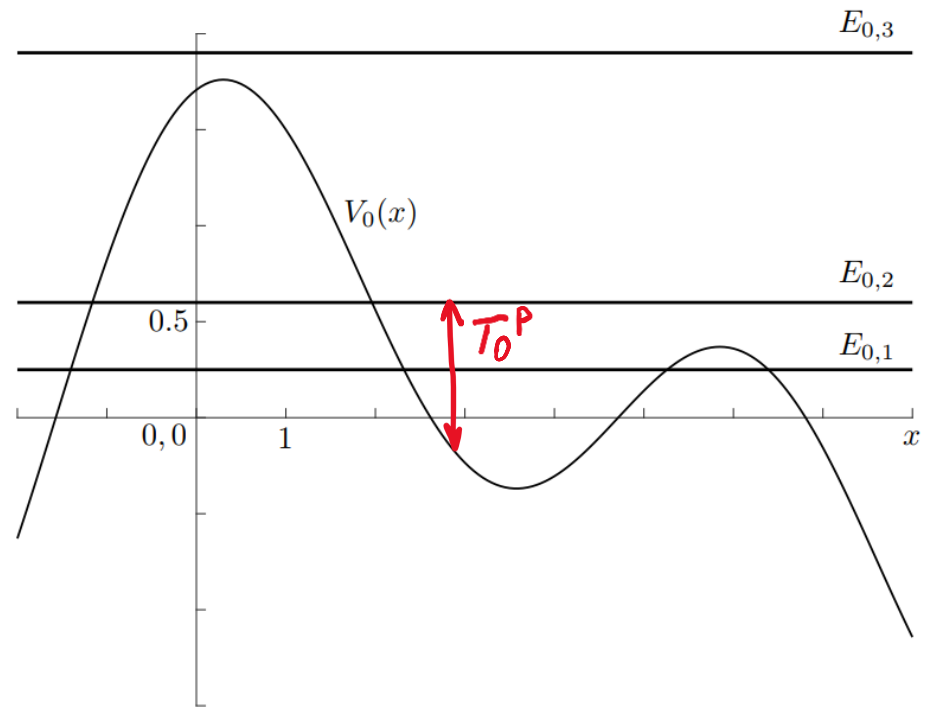
Analysis of motion when E_0^P is conserved

➤ The conservation of mechanical energy is specially interesting in 1D problems

- We plot $V_0(x)$ and the line $V_0 = E_0^P$
- The difference between V_0 and E_0^P gives the value of T_0^P , and hence the speed v_0^P

$$T_0^P = \frac{1}{2}(v_0^P)^2 = E_0^P - V_0$$

- T_0^P must be ≥ 0 !!
- Given the value of E_0^P , which can be computed from the initial conditions, we can determine *allowed* and *forbidden* regions of space



An important skill for the student:
given the function $V_0(x)$, sketch the
function and analytically find
maxima/minima, asymptotes, etc

Equilibrium configurations

- We say a configuration $\mathbf{r}_0^P = \mathbf{r}_0^*$ is an *equilibrium* one if, when we leave the particle in it with zero velocity, it remains at rest indefinitely. This means that the particle acceleration must be zero:

$$\mathbf{a}_0^P = \mathbf{0}$$

- From Newton's second law, this occurs iff the sum of all forces acting on the particle is zero:

$$\mathbf{F}(\mathbf{r}_0^*) = \mathbf{0}$$

- Equilibrium configurations are *stable* if, when we perturb the position or the velocity of the particle slightly with respect to that configuration, the particle does not depart far from it.

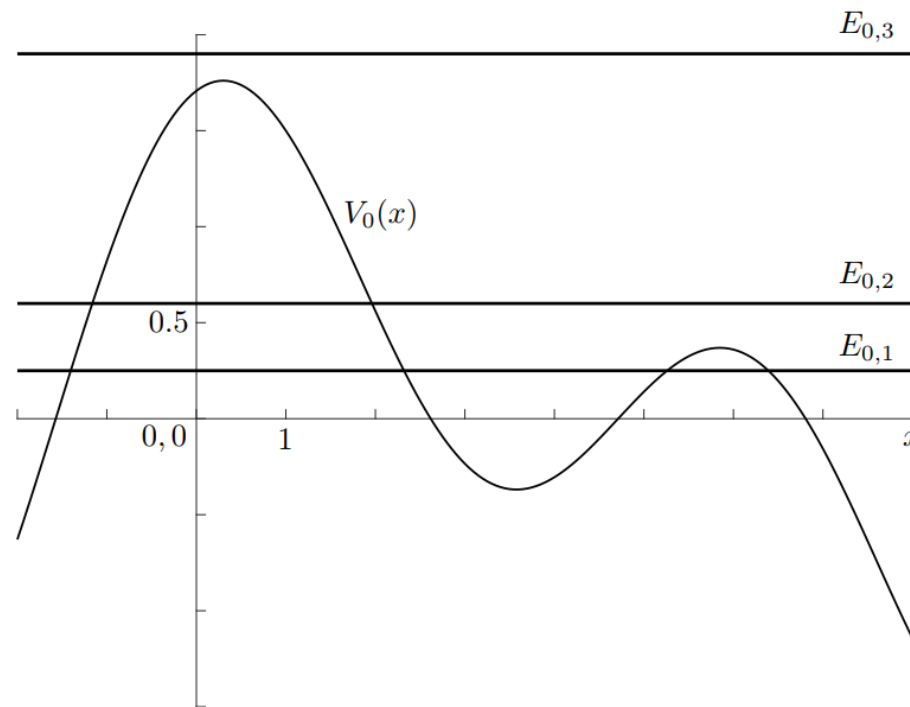
- $\mathbf{r}_0^*$ is stable iff $\forall \varepsilon_r, \varepsilon_v > 0, \exists \delta_r, \delta_v > 0$ such that if we perturb P initially with $|\mathbf{r}_0^P(0) - \mathbf{r}_0^*| < \delta_r$ and $|\mathbf{v}_0^P(0)| < \delta_v$, then $|\mathbf{r}_0^P(t) - \mathbf{r}_0^*| < \varepsilon_r$ and $|\mathbf{v}_0^P(t)| < \varepsilon_v$ for all $t > 0$.

- Otherwise, the equilibrium configuration is *unstable*.



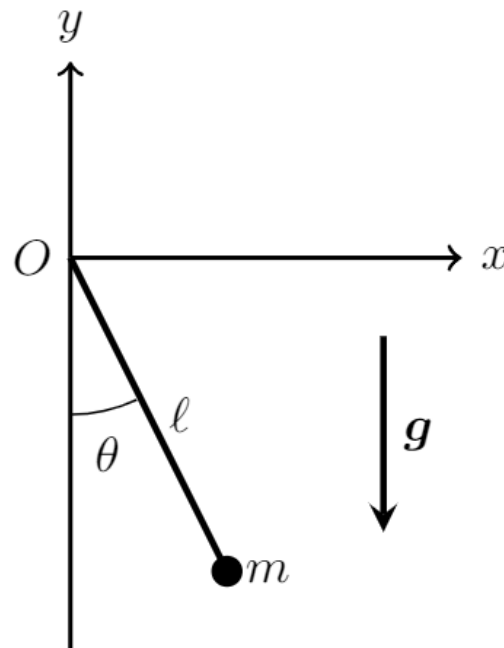
Equilibrium configurations

- In conservative, one-dimensional problems, we may deduce the equilibrium configurations and their stability from $V_0(x)$:
- Equilibrium: $F = -dV_0/dx = 0$. These are the *critical points* of V_0
 - Stable equilibrium configurations: $dF/dx = -d^2V_0/dx^2 < 0$ (minima)



Example: Simple pendulum

- Kinematics and Dynamics of this simple system
- Discuss conservation of energy
- Linearize the equations of motion and discuss



Thank you!

